

1. General Description

The EMS3550 is a high-performance four-pole single-throw (4-PST) analog switch with wide V_{DD} range. The key feature of the EMS3550 is the depletion-mode isolation technology. That allows the device to conduct signals when there is no V_{DD} available and to isolate the signals when V_{DD} is present. In addition, select pins allow the internal oscillator frequency to be adjusted between 790 kHz and 1.12 MHz in 110 kHz steps when V_{DD} is present. This feature can improve electromagnetic interference (EMI).

2. Features and Benefits

- Four SPST depletion switches
- Normally closed when $V_{DD} < 0.2V$
- V_{SW} : -3 V to 4.7 V
- R_{ON} : 1.2 Ω (typ.)
- $R_{FLAT(ON)}$: 14 m Ω (typ.)
- THD+N: -104 dB

3. Applications

- Smart phones, Tablets, laptops
- Bluetooth headsets, Mobile accessories

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4. Ordering Information

Table 1. Ordering information

Type number	Topside marking	Package		
		Name	Description	Quantity
EMS3550UD	550Q YWW	QFN-16L	QFN package, 16 pins; 3 mm x 3 mm; 0.75 mm (Max) height	5000
EMS3550YFC		WLCSP-12	WCSP package, 12 balls; 1.45 mm x 1.86mm; 0.625 mm(Max)height	3000

5. Function Diagram

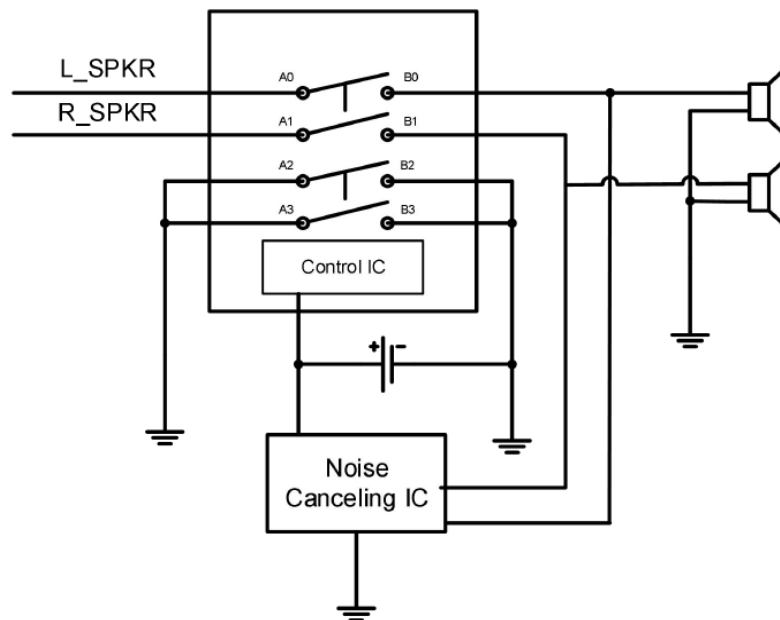
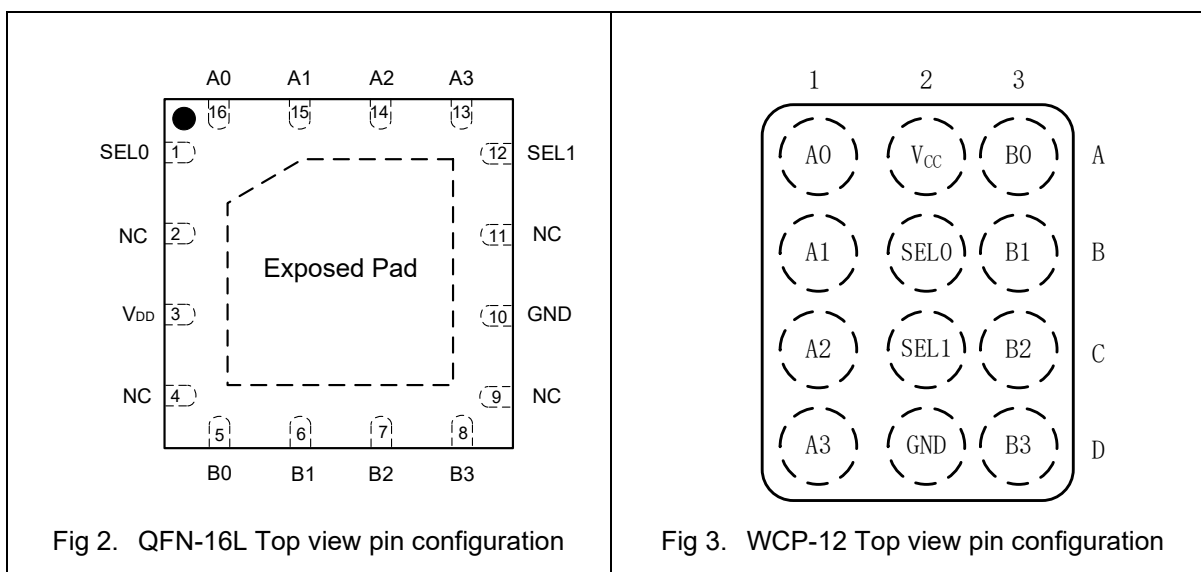


Fig 1. Typical application circuit

6. Pinning Information

6.1. Pin map



6.2. Pin description

Table 2. Pin description

Symbol	Pin	Type	Description
A0	16	I/O	A Port
A1	15	I/O	A Port
A2	14	I/O	A Port
A3	13	I/O	A Port
V _{DD}	3	Supply/Control	Isolation Circuit Supply Voltage
SEL0	1	I	Oscillator frequency control (see Table 4). Used to shift the electromagnetic interference (EMI) signature to meet the customer specifications.
SEL1	12	I	
GND	10	Ground	System Ground
B0	5	I/O	B Port
B1	6	I/O	B Port
B2	7	I/O	B Port
B3	8	I/O	B Port

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7. Functional Description

Table 3. Function table

V_{DD}	Function
0 V to 0.2 V	B0 - B3 connect to A0 - A3
2.5 V to 5 V	B0 - B3 disconnect to A0 - A3

Table 4. Oscillator Frequency Step Logic

SEL1	SEL0	Frequency(Typ.)
Low	Low	790 kHz
Low	High	900 kHz
High	Low	1010 kHz
High	High	1120 kHz

8. Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only.

Table 5. Absolute Maximum Ratings

In accordance with the Absolute Maximum Rating System (IEC 60134). Voltages are referenced to GND.

Symbol	Parameter	Min	Max	Unit
V_{CC}	Supply voltage	-0.5	5.5	V
V_{SW}	DC switch I/O voltage (switch conducting)	-3.5	5	V
	DC switch I/O voltage (switch isolated)	-3.5	5	V
I_{SW}	Maximum peak switch I/O current+ -pulsed at 1ms duration,<10% duty cycle		500	mA
T_{stg}	Storage Temperature	-65	150	°C
ESD	HBM, JEDEC: JESD22-A114	-4	4	kV

9. Recommended Operating Conditions

The Recommended Operating Conditions table defines the conditions for actual device operation. Recommended operating conditions are specified to ensure optimal performance to the datasheet specifications. EnergyMath does not recommend exceeding them or designing to Absolute Maximum Ratings.

Table 6. Recommended Operating Conditions

Symbol	Parameter	Min	Max	Unit
$V_{DD(OFF)}$	Supply voltage, Isolating	2.5	5.0	V
$V_{DD(ON)}$	Supply voltage, Conducting	0	0.2	V
$V_{SW(OFF)}$	Isolating ($V_{DD} = 2.5$ V)	-1.5	2.5	V
	Isolating ($V_{DD} = 3.3$ V)	-2.5	3.3	V
	Isolating ($V_{DD} = 5.0$ V)	-3.0	4.7	V
$V_{SW(ON)}$	Conducting	-3.0	4.7	V
T_A	Ambient operating Temperature	-40	85	°C

10. Electrical Characteristics

Table 7. Static characteristics
 $V_{DD} = 0V$, $T_A = +25^{\circ}C$, unless otherwise specified.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
DC Electrical Characteristics						
I_{OFF}	Switch off leakage current	$V_{DD} = 5V, B_n = GND$	$A_x = 4.7V$		0.01	μA
			$A_x = -1.5V$		0.7	μA
			$A_x = 1.5V$		-0.3	μA
		$V_{DD} = 3.3V, B_n = GND$	$A_x = -1.5V$		0.1	μA
			$A_x = 1.5V$		0.1	μA
R_{ON}	Switch on resistance	$I_{SW} = 50mA, V_{SW} = -3V \text{ to } 3V$		1.2		Ω
$R_{FLAT(ON)}$	On resistance flatness	$I_{SW} = 50mA, V_{SW} = -3V \text{ to } 3V$		14		m Ω
C_{ON}	On capacitance	$V_{SW} = 0dBm, f = 1MHz$		42		pF
C_{OFF}	Off capacitance	$V_{SW} = 0dBm, f = 1MHz$		31		pF
AC Electronics Characteristics						
THD+N	Total harmonic distortion plus noise	$V_{SW} = 0.7V_{RMS}, f = 1kHz @ R_L = 32\Omega$		-104		dB
OIRR	Off isolation rejection ratio	$V_{SW} = 1V_{RMS} @ R_L = 32\Omega$	$f = 1kHz$		-100	dB
			$f = 20kHz$		-85	dB
BW	Band Width	$R_L = 50\Omega, C_L = 0pF$		21		MHz
Supply currents						
I_{DD}	Quiescent current	$V_{DD} = 2.5V, SEL0 = SEL1 = 0V$		67		μA
		$V_{DD} = 3.3V, SEL0 = SEL1 = 0V$		92		μA
		$V_{DD} = 5.0V, SEL0 = SEL1 = 0V$		158		μA
I_{DIS}	Disable current	$V_{DD} \leq 0.2V$		0.1		μA
Control logic (Select pins)						
V_{IH}	Input voltage high		1.6			V
V_{IL}	Input voltage low				0.4	V

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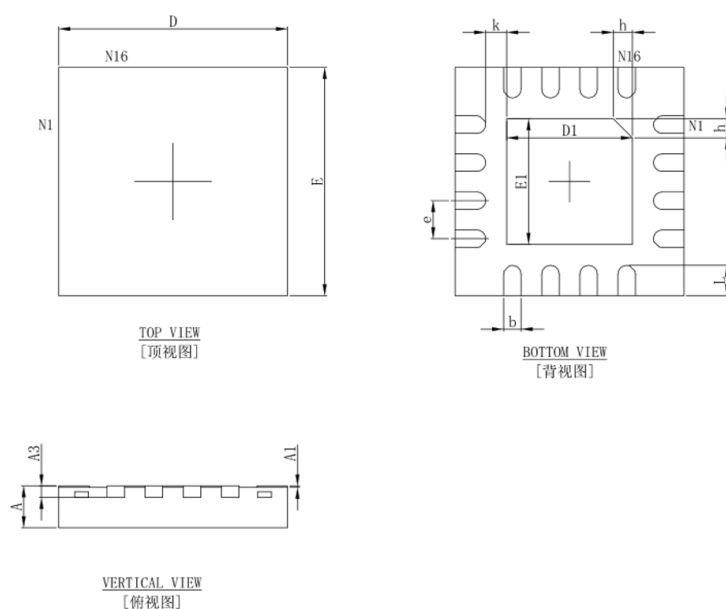
Symbol	Parameter	Conditions	Min	Typ	Max	Unit
Switching characteristics						
t_{ON}	Switch turn-on Time	$R_L = 1\text{ k}\Omega$, $C_L = 10\text{ pF}$, $V_{DD} = 2.5\text{ V to }0\text{ V}$, $V_{SW} = 1.5\text{ V}$ $SEL0=SEL1=0V$		40		μs
t_{OFF}	Switch turn-off Time	$R_L = 1\text{ k}\Omega$, $C_L = 10\text{ pF}$, $V_{DD} = 0\text{ V to }2.5\text{ V}$, $V_{SW} = 1.5\text{ V}$ $SEL0=SEL1=0V$		400		μs

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11. Package Outline

QFN 3x3-16L

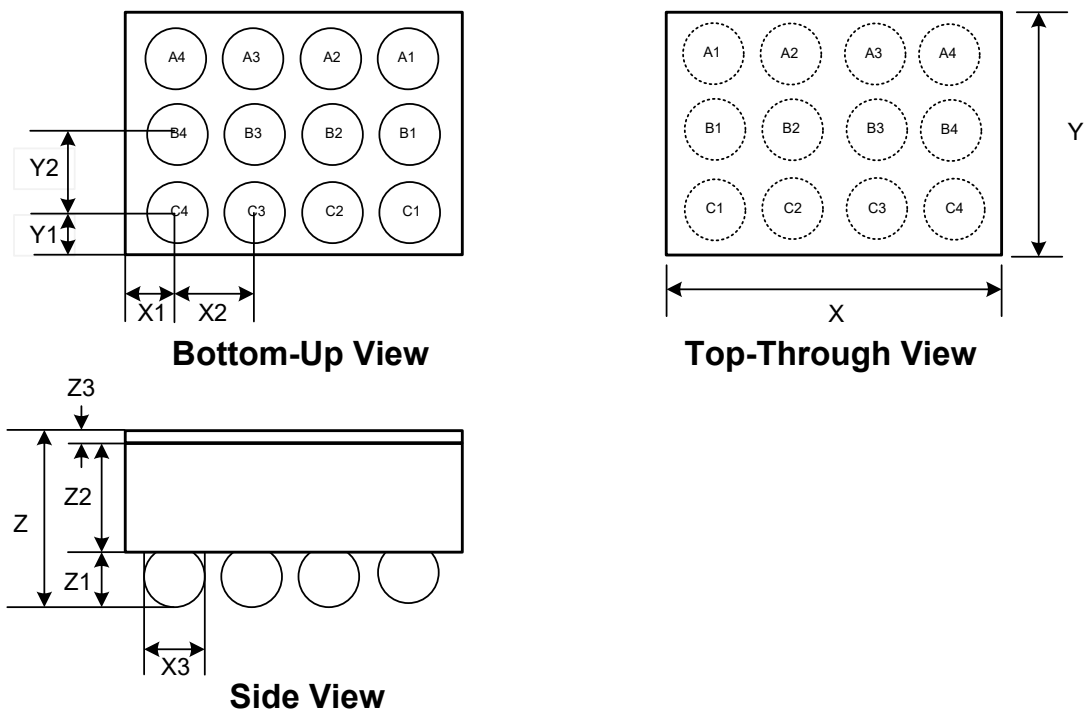


SYMBOL	MIN	NOM	MAX
A	0.700	0.750	0.800
A1	0.000	0.020	0.050
A3	0.203 REF		
b	0.180	0.230	0.280
D	2.900	3.000	3.100
E	2.900	3.000	3.100
e	0.500 BSC.		
D1	1.550	1.650	1.750
E1	1.550	1.650	1.750
L	0.300	0.400	0.500
K	0.200 MIN.		
h	0.250 REF.		

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WLCSP-12



Symbol	Dimensions In Millimeter		
	Min.	Typ.	Max.
X	1.83	1.86	1.89
Y	1.42	1.45	1.48
X1		0.33	
X2		0.40	
X3	0.21	0.23	0.25
Y1		0.325	
Y2		0.40	
Z	0.525	0.575	0.625
Z1	0.165	0.185	0.205
Z2	0.340	0.365	0.390
Z3	0.020	0.025	0.030

12. Revision History

Table 8. Revision history

Document ID	Release Date	Data sheet status	Change notice	Supersedes
EMS3550 Rev. 1.0	Mar 10, 2026	Product datasheet		